Intrusion Detection Systems, Malware Detection, and Firewalls

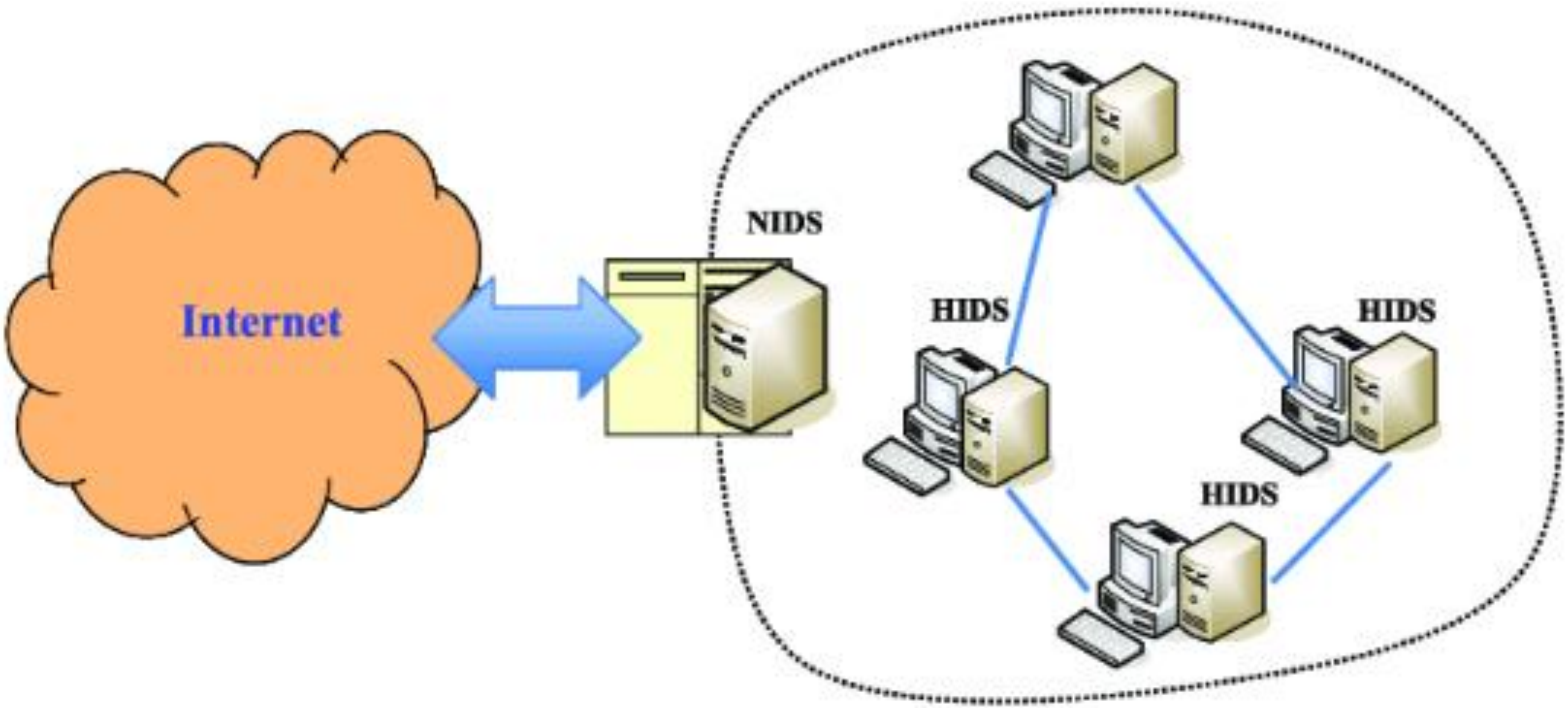
Intrusion Detection Systems (IDS)

- An Intrusion Detection System (IDS) is a security system designed
  - to detect and respond to potential threats, attacks, or suspicious activities in a network or system.
  - IDS monitors traffic, logs, and behavior patterns to identify malicious activities.
- Objectives of IDS:
  - Detect unauthorized access to network resources.
  - Monitor network traffic for signs of malicious activity.
  - Generate alerts for potential security breaches.
  - Log events for later investigation and analysis.
  - Prevent or limit damage by taking action based on detected threats.

Types of IDS:

- There are two primary types of IDS
- Network-based IDS (NIDS):
  - Monitors network traffic to detect suspicious activities.
  - Placed at strategic points (like network perimeters) to capture data.
  - Ideal for detecting attacks like Denial of Service (DoS), port scanning, and buffer overflow attacks.
- Host-based IDS (HIDS):
  - Monitors activities and events on a single host (e.g., server or workstation).
  - Looks for changes in system files, logs, or unusual application behavior.
  - Effective for detecting attacks that bypass network defenses (e.g., rootkits, privilege escalation).

Block Diagram placing NIDs and HIDS



IDS Detection Techniques

- Signature-based Detection:
  - Compares traffic patterns to a database of known attack signatures.
  - Effective for known threats, but cannot detect new or unknown attacks (zero-day vulnerabilities).
  - Example: Virus signature detection.
- Anomaly-based Detection:
  - Establishes a baseline of normal behavior for users or networks.
  - Detects deviations from this baseline (e.g., a sudden spike in traffic or abnormal system behavior).
  - Can identify unknown threats but may have false positives.

Hybrid Detection

- Combines both signature-based and anomaly-based methods.
- Provides better coverage by detecting known and unknown threats.
- Often used in modern IDS solutions.

IDS Response Mechanisms

- Alerting: Sends notifications when an attack is detected.
- Logging: Records the event for later analysis.
- Active Responses: In advanced IDS systems, automated actions such as blocking the attack or isolating the affected system can be performed.

Malware Detection and Prevention

- Malware (short for malicious software) refers to any software or code that is intentionally designed to
  - damage,
  - disrupt,
  - or gain unauthorized access to a computer system or network.
- Types of malware include viruses, worms, trojans, ransomware, and spyware.
- Objectives of Malware Detection:
  - Identify harmful or suspicious software before it causes damage.
  - Prevent infection by blocking malware from entering the system.
  - Quarantine or remove malware once detected.

Malware Detection Techniques

- Signature-based Detection:
  - Relies on known malware signatures (similar to IDS).
  - Detects malware by comparing files and behaviors with a signature database.
  - Effective for known malware but ineffective for zero-day attacks or new variants.
- Heuristic-based Detection:
  - Analyzes the behavior of software or code to identify suspicious activities.
  - Looks for common patterns of malware behavior, such as unauthorized file modifications or excessive resource consumption.
  - Can detect new or unknown malware, but may produce more false positives.

Malware Detection Techniques

- Behavioral-based Detection:
  - Focuses on monitoring the behavior of programs during execution (e.g., file system changes, network activity).
  - Identifies malicious activity by observing suspicious actions rather than static signatures.
- Sandbox Analysis:
  - Runs suspicious programs in an isolated environment (sandbox) to observe their behavior.
  - If the program behaves maliciously (e.g., tries to encrypt files or send sensitive data), it is flagged as malware.

Malware Prevention Techniques

- Antivirus/Antimalware Software:
  - Scans files for known malware signatures.
  - Uses heuristic or behavioral analysis to detect malware.
  - Regular updates to ensure protection against newly discovered threats.
- Firewalls:
  - Prevents unauthorized access to systems that could allow malware to enter.
  - Often includes packet filtering, deep packet inspection, and application-layer filtering.

Malware Prevention Techniques

- User Education:
  - Avoid suspicious links and email attachments.
  - Ensure regular software updates and patch management.
- Secure Configurations:
  - Disable unnecessary services to reduce vulnerabilities.
  - Regular updates to operating systems and applications.

Firewalls

- A firewall is a network security system that monitors and controls incoming and outgoing traffic based on predefined security rules.
- It serves as a barrier between a trusted internal network and untrusted external networks (like the Internet).
- Objectives of Firewalls:
  - Protect networks from unauthorized access.
  - Block malicious traffic or traffic from suspicious sources.
  - Enforce security policies for traffic entering or leaving the network.

Types of Firewalls

- Packet Filtering Firewalls:
  - Basic firewall type that inspects individual packets based on predefined rules.
  - Filters traffic based on IP address, port number, and protocol.
  - Example: A rule might allow HTTP traffic (port 80) and block all incoming traffic on port 23 (Telnet).
- Stateful Inspection Firewalls:
  - Tracks active connections and ensures that incoming packets are part of a valid session.
  - More secure than packet filtering as it maintains state tables to inspect traffic more thoroughly.

Types of Firewalls

- Proxy Firewalls:
  - Acts as an intermediary between clients and servers (application-level firewall).
  - Can filter and inspect traffic at the application layer.
  - Typically used to block malicious websites or applications.
- Next-Generation Firewalls (NGFW):
  - Includes features of stateful inspection, packet filtering, and application-level filtering.
  - Deep packet inspection (DPI) and intrusion prevention systems (IPS) are integrated.
  - Provides detailed traffic analysis and advanced security policies.

- Web Application Firewalls (WAF):
    - Protects web applications from attacks like SQL injection, XSS, and other application layer attacks.
    - Can filter and monitor HTTP traffic to ensure only legitimate requests reach web servers.

Firewall Operation

- Packet Filtering:
  - Inspects each packet of data and decides whether to allow or block it based on rules.
- Stateful Inspection:
  - Tracks the state of active connections and filters packets accordingly.
- Proxying:
  - Intercepts requests and responses between clients and servers to ensure they comply with security policies.